Quantum Donuts

QIF - Challenge 1

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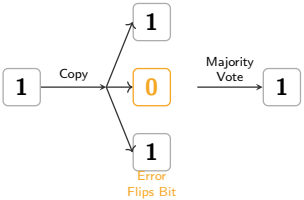
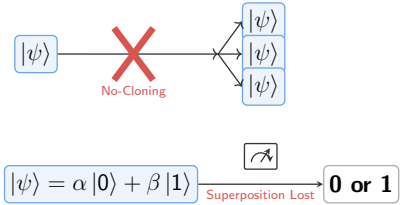
The Quantum Error Correction Challenge

- **Quantum Fragility:** Quantum states are highly susceptible to noise and imperfect controls.
- **The Classical Solution:** Classical computers protect data via redundancy (e.g., copying a bit 3 times and taking a majority vote).
- **The Quantum Roadblocks:**
  1. **No-Cloning Theorem:** Physically impossible to create an identical copy of a qubit.
  2. **Measurement Collapse:** Measuring a data qubit to check for errors destroys its superposition.

Classical Redundancy (Works)

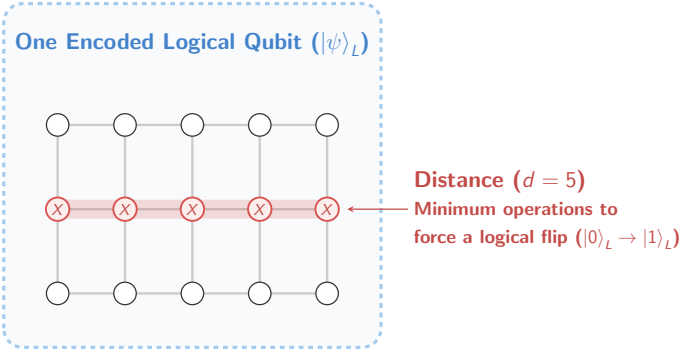


Quantum Roadblocks (Fails)



Logical qubits vs Physical

- **Logical Qubits:** Composition of entangled physical qubits, highly stable, abstract states $|0\rangle_L, |1\rangle_L$.
- **Physical Qubits:** The (noisy) hardware qubits inside the quantum computer.
- **Distance (d):** The minimum amount of physical gates/errors that change the logical state.
- **Correct ($d/2$) errors**.

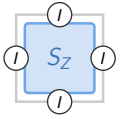


The Quantum Solution: Anti-Commutation

- **Stabilizers:** Parity operators (e.g. $S = Z_1 Z_2 Z_3 Z_4$) such that $S\,|\psi\rangle_L = |\psi\rangle_L$.
- **Stabilizers make Identity:** $\prod S = \mathbb{I}$
- **Anti-Commutation:** If Pauli X error occurs, it *anti-commutes* with Z-stabilizers

$$Z_1 Z_2 Z_3 Z_4 X_1\,|\psi\rangle_L = Z_1 X_1 Z_2 Z_3 Z_4\,|\psi\rangle_L = -X_1 Z_1 Z_2 Z_3 Z_4\,|\psi\rangle_L = -X_1\,|\psi\rangle_L\,.$$

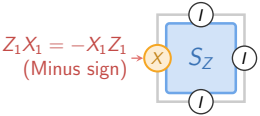
No Error (Commutes)



Outcome: $+1$

$S_Z\,|\psi\rangle = (+1)\,|\psi\rangle$

X_1 Error (Anti-Commutes)



Outcome: -1 **(Error!)**

$S_Z(X_1\,|\psi\rangle) = (-1)X_1\,|\psi\rangle$

$S_Z = Z \otimes Z \otimes Z \otimes Z$ **vs** $E = X \otimes I \otimes I \otimes I$

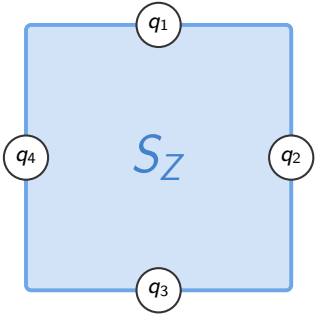
Stabilizers (Plaquette and Cross)

Phase-flip Stabilizer (S_Z)

Parity of 4 edge qubits forms a **face**.

$$S_Z = Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4$$

Detects X (bit-flip) errors.



Bit-flip Stabilizer (S_X)

Parity of 4 edge qubits meeting at a **vertex**.

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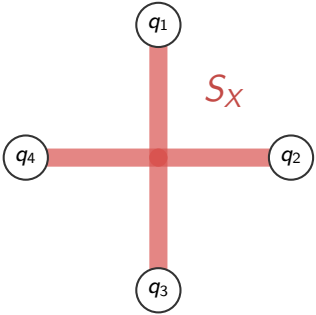


Toric code

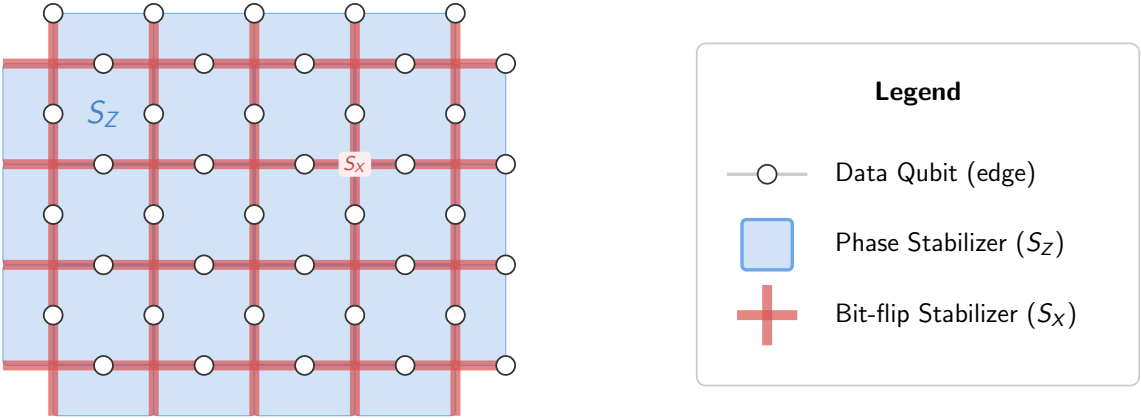
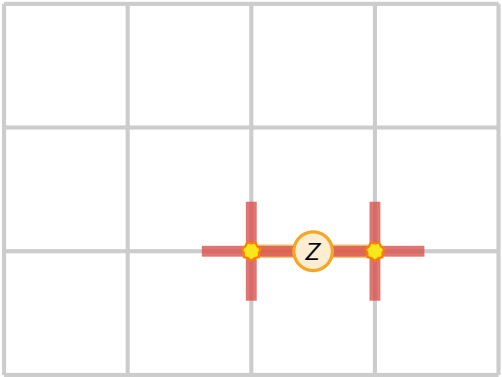


Figure: Toric Code Lattice with **periodic boundaries**.

Syndromes appear in pairs

- **Single Errors:** A single phase error (Z) on a physical data qubit anti-commutes with its adjacent X-stabilizers, triggering **two** syndrome flags.



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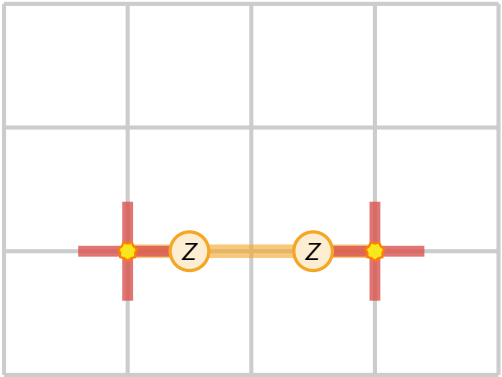
- **The Result:** The intermediate flags perfectly cancel out! We only ever observe syndromes at the **endpoints** of an error path.



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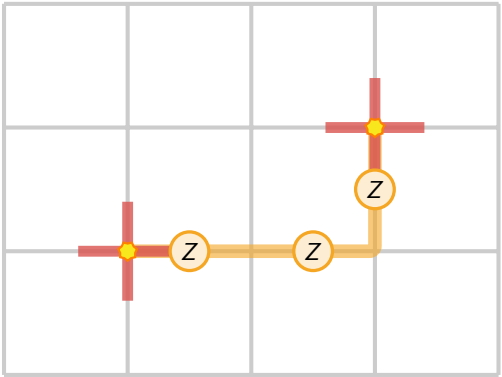

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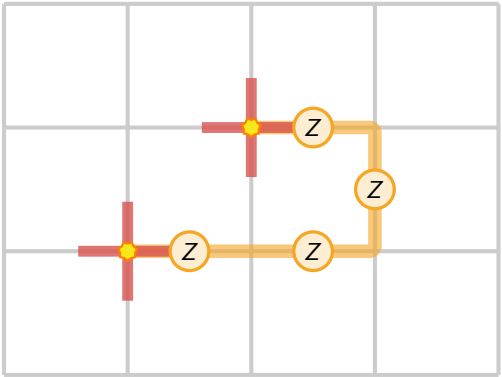
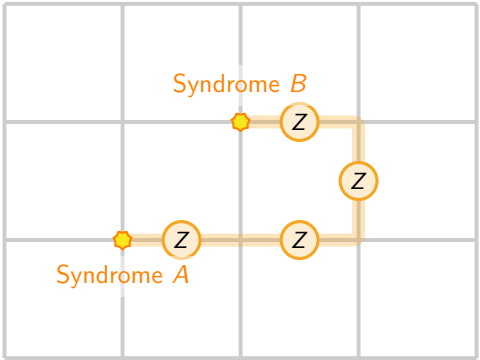


Inferring error "path"

- **The Challenge:** We only observe the syndromes (endpoints).
- **The Goal:** Reconstruct the most likely chain of Z errors between them.
- **Key insight:** If our inferred path and the real path form a loop, the state $|\psi\rangle_L$ is protected.



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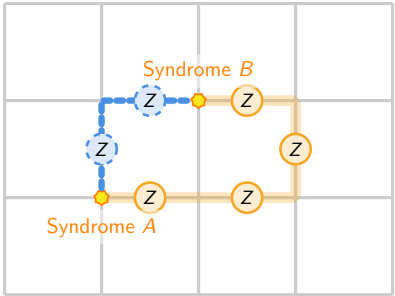


Inferred Correction: Shortest Manhattan Distance

Are all loops good?

Trivial vs Non-Trivial Loops

- **Trivial Loops (Local):**
  - Can be continuously shrunk to a single point.
  - **Result:** Error chains on these loops do *not* alter the logical state.
- **Non-Trivial Loops (Global):**
  - Wraps around the lattice topology.
  - Error chains on these loops *do* alter the logical state!

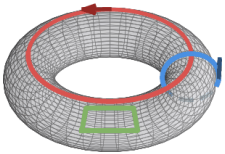
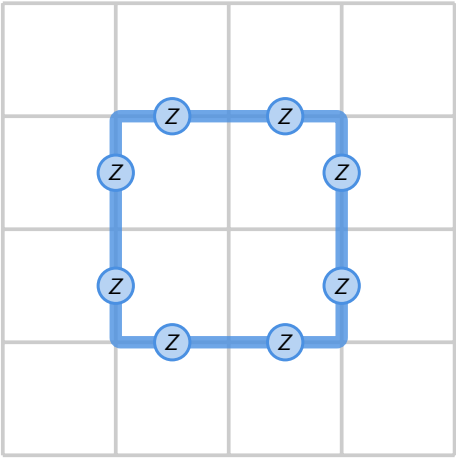


Figure: Possible Torus loops

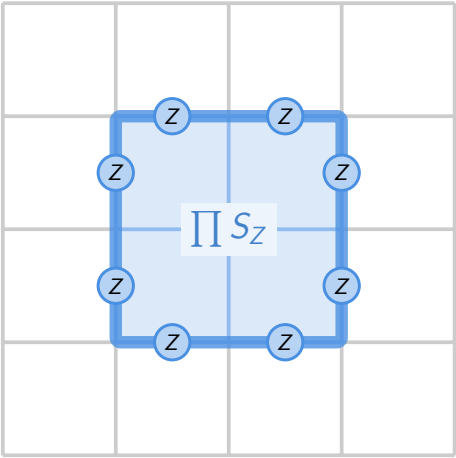
Trivial loops are stabilizers

- **Trivially Closed Paths = No Flags:** If a chain of errors forms a trivial loop, it has no endpoints. Therefore, it triggers exactly zero syndrome flags.



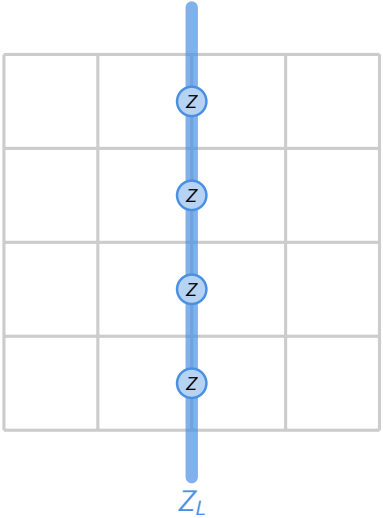
Trivial loops are stabilizers

- **Trivially Closed Paths = No Flags:** If a chain of errors forms a trivial loop, it has no endpoints. Therefore, it triggers exactly zero syndrome flags.
- **Product of Stabilizers:** Any trivial loop equals the product of the stabilizers enclosed in it (which is itself a stabilizer).
- **Harmless Operations:** $S\,|\psi\rangle_L = |\psi\rangle_L$.



Non-trivial loops are logical gates

- **Spanning the Lattice:** If an error path crosses from one boundary to the opposite boundary, it acts as a non-trivial loop.



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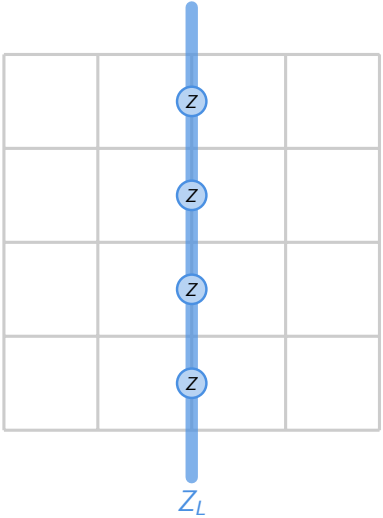

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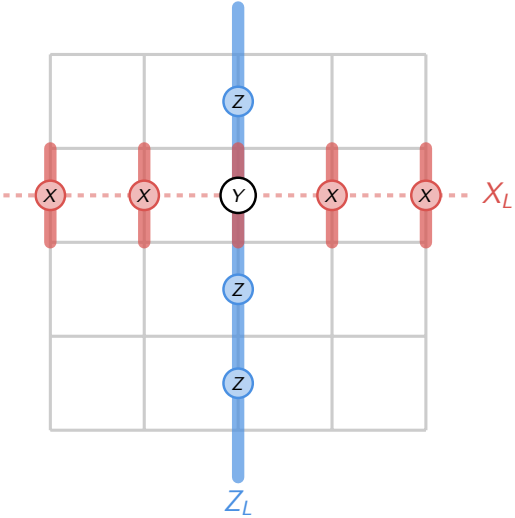
- **Spanning the Lattice:** If an error path crosses from one boundary to the opposite boundary, it acts as a non-trivial loop.
- **Undetectable Error:** It has no endpoints just like trivial loops, so it triggers **zero syndrome flags**.
- **Logical Corruption:** They change the global state (e.g. X_L flips $|0\rangle_L \to |1\rangle_L$).



Why does the Toric Code encode 2 qubits?

- **Topology = Capacity:** The number of logical qubits encoded (k) equals the number of independent "holes" in the surface. A torus has **2** distinct ways to draw non-trivial loops.

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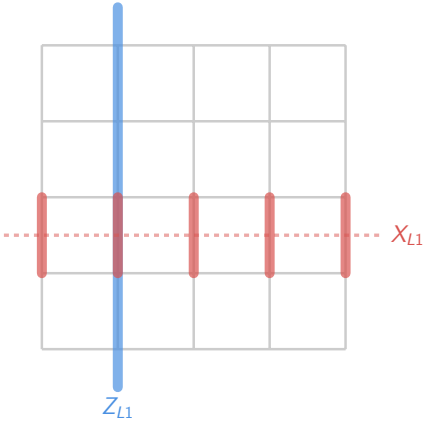
  - Encoded entirely in the **vertical-edge** physical qubits.
  - Z_{L1} loops vertically (primal path).
  - X_{L1} loops horizontally (dual path).



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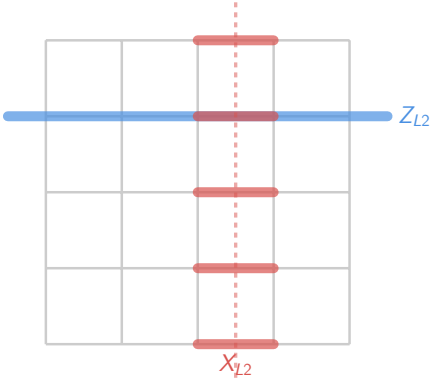
- **Logical Qubit 2:**
  - Encoded entirely in the **horizontal-edge** physical qubits.
  - Z_{L2} loops horizontally (primal path).
  - X_{L2} loops vertically (dual path).



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- **Logical Qubit 2:**
  - Encoded entirely in the **horizontal-edge** physical qubits.
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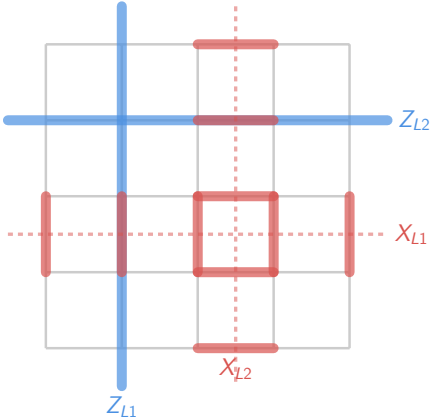
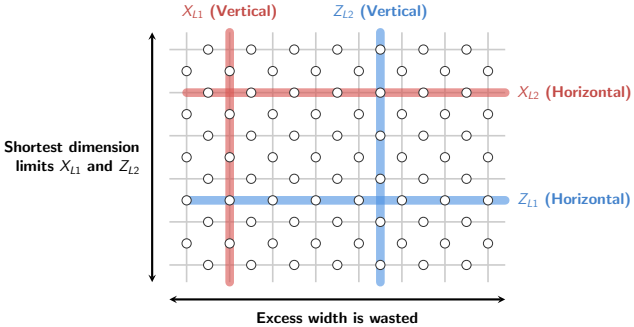
  - X_{L2} loops vertically (dual path).
- **Perfect Independence:** X_{L1}, X_{L2} never overlap. They commute perfectly!



What about unbalanced errors?

Unbalanced errors

- **Biased Noise:** If phase-flip errors are more likely than bit-flip errors ($p_Z \gg p_X$).
- **Orthogonal Operators:** Logical operators for different logical qubits span orthogonally. Here X_{L1} is vertical, while X_{L2} is horizontal.
- **Symmetry:** X_L strings exist in *both* directions: lattice width = height.
- **Result:** The lattice is forced to be a massive square of $\max(d_X, d_Z)$.



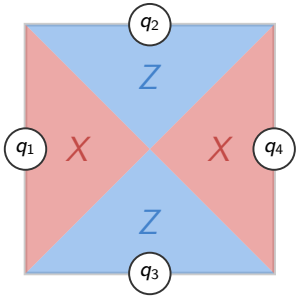
Asymmetric Stabilizers

Tailored Face

Measures the $XZZX$ parity of 4 edge qubits forming a closed face.

$$S_{face} = X_1 \otimes Z_2 \otimes Z_3 \otimes X_4$$

Horizontal edges apply X, vertical edges apply Z.



Tailored Vertex

Measures the $ZXXZ$ parity of 4 edge qubits meeting at a vertex.

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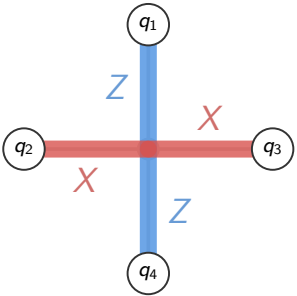
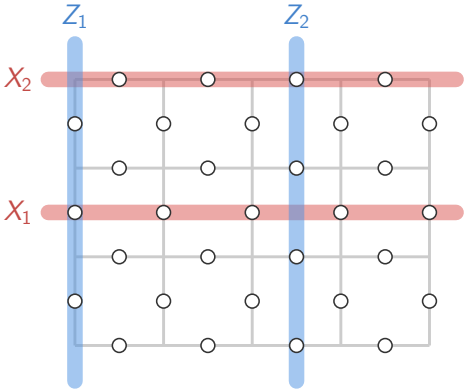


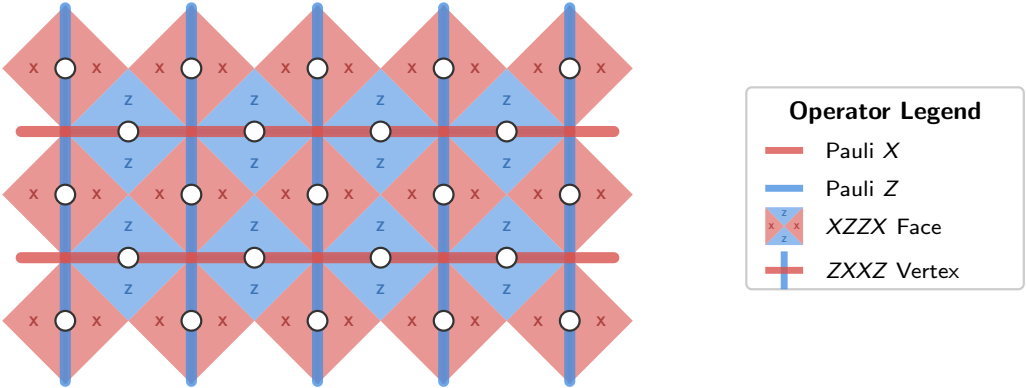
Asymmetrical Logical Operators

- **Directional Asymmetry:** The local basis rotation changes the logical operators.
- **Intersections:** Both X_1Z_1 and Z_2X_2 intersect on *one* qubit (i.e. they anti-commute)
- **No intersections:** Neither X_1Z_2 nor Z_1X_2 intersect on any qubit (i.e. they commute)



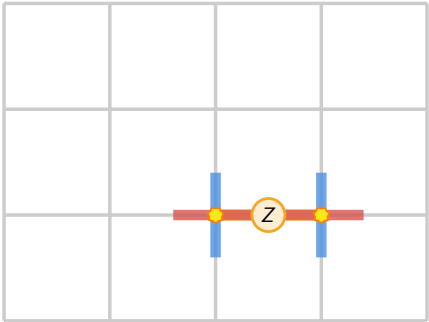
Asymmetric Surface Code

- **Biased Noise Adaptation:** By applying a local basis rotation ($X \leftrightarrow Z$ on vertical edges), we transform standard $XXXX, ZZZZ$ stabilizers into mixed $XZZX, ZXXZ$ operators.
- **Asymmetry:** A $d_x \times d_z$ rectangular patch extends the distance for one error type (e.g. $p_x \gg p_z$).



Tailored Error Chains

- **Orientation-Dependent Chains:** In a tailored $XZZX$ code, a continuous error chain alternates between X and Z depending on the edge orientation.
- **Turning Corners:** A Z error on a horizontal edge triggers the X arms of the vertices. To turn a corner, an X error must occur on the vertical edge to trigger the Z arms.



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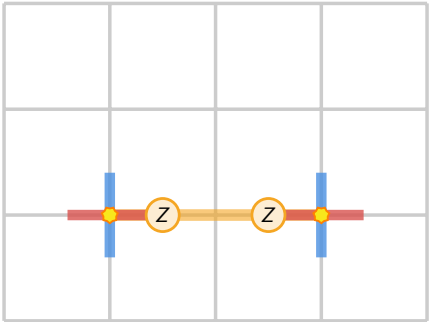

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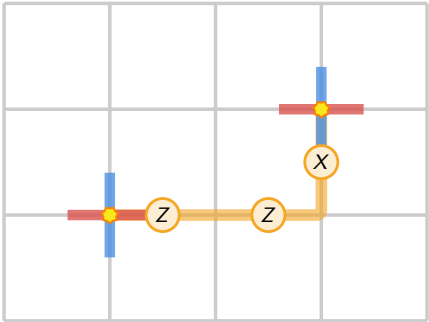

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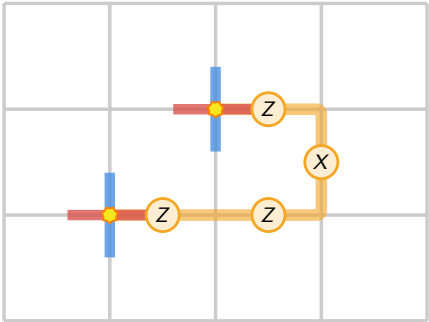
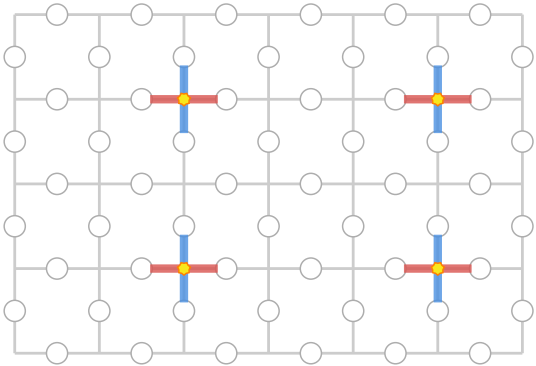


Biased Syndrome Matching

- **Ambiguous Syndromes:** A square of 4 flags can be paired in two ways: vertical or horizontal.
- **Anisotropic Edge Weights:** Assume $\mathbf{p_Z} \gg \mathbf{p_X}$ (e.g. $0.1 \gg 10^{-5}$)
- **Asymmetric decoding:** Decoder *prefers* horizontal links (Z errors) to vertical links (X errors).

Ambiguity: 4 Flags Detected



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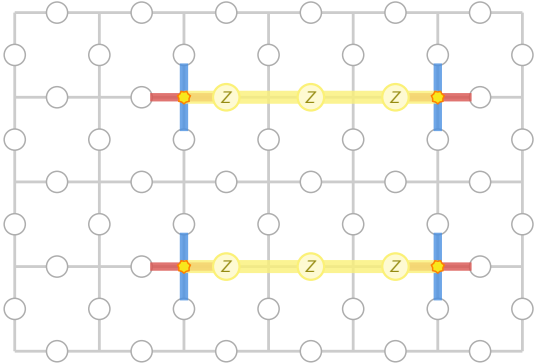
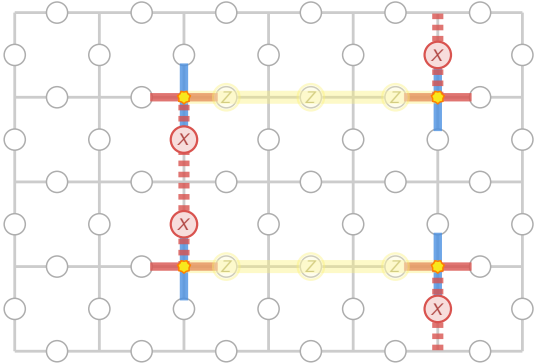
Actual Error Event: High-Probability Horizontal Z Strings (Weight 3)



Biased Syndrome Matching

- **Ambiguous Syndromes:** A square of 4 flags can be paired in two ways: vertical or horizontal.
- **Anisotropic Edge Weights:** Assume $\mathbf{p_Z} \gg \mathbf{p_X}$ (e.g. $0.1 \gg 10^{-5}$)
- **Asymmetric decoding:** Decoder *prefers* horizontal links (Z errors) to vertical links (X errors).

(Standard Decoder) Shorter Vertical X Pairing (Weight 2) $\implies$ Logical Error!



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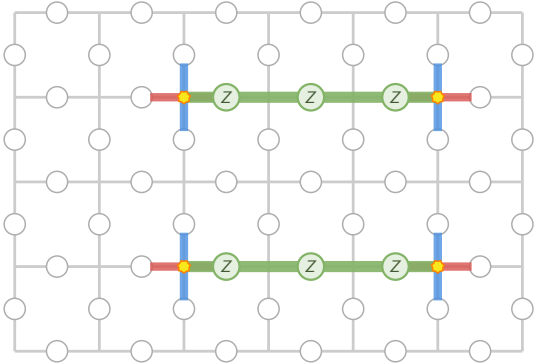
(Biased Decoder) Longer Horizontal Z Pairing (Weight 3) $\implies$ Safe / Exact Match

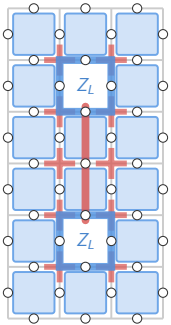


Including Extra Logical Qubits

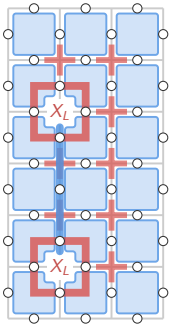
Creating Logical Qubits via Defects

- **Defect Qubits:** Turning off stabilizers frees up degrees of freedom for extra logical qubits.
- **Double Z-cut (Left):** Creates a qubit where X_L connects the holes, Z_L encloses them.
- **Double X-cut (Right):** Creates a qubit where Z_L connects the holes, X_L encloses them.
- *Source: Fowler et al. 2012*

Double Z-cut Qubit



Double X-cut Qubit



Operator Legend

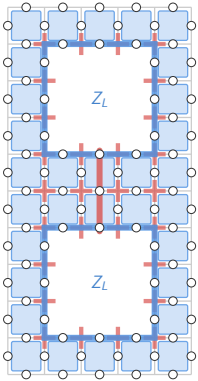
Pauli X

Pauli Z

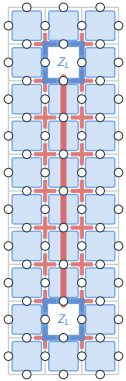
Scaling Up: Larger Logical Qubits

- **Fault Tolerance:** The distance d of a defect qubit is bounded by the hole perimeter and spacing.
- **Multi-Cell Holes:** Turning off a larger block of stabilizers expands the defect.
- **Extended Operators:** A longer physical error chain is required to disrupt the logical state.

Closer, Larger Holes
($d_X = 3, d_Z = 12$)



Farther, Smaller Holes
($d_X = 7, d_Z = 4$)



Operator Legend

Pauli X

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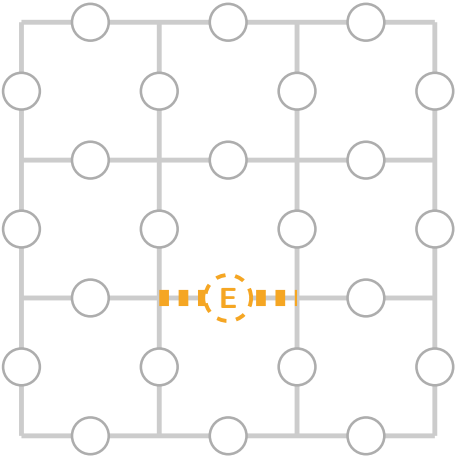
What about Erasure errors?

Recovering from Erasures

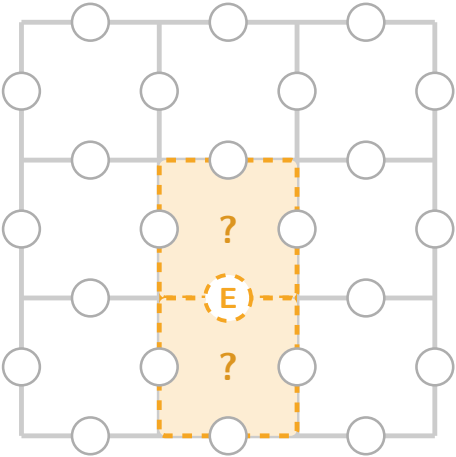
- **Erasure vs. Pauli Error:** The qubit is completely lost, but we know *where* it happened.



Recovering from Erasures

- **Erasure vs. Pauli Error:** The qubit is completely lost, but we know *where* it happened.
- **Blinded Stabilizers:** Adjacent X and Z stabilizers are randomized.

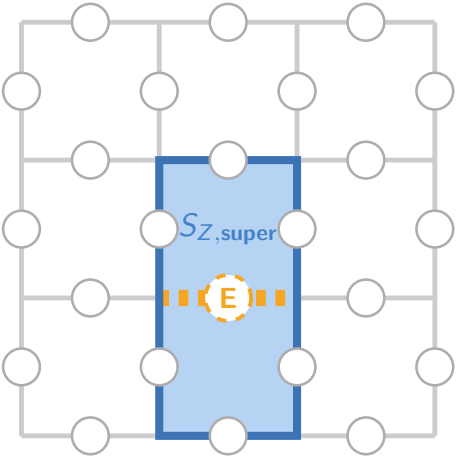
Adjacent Stabilizers Blinded



Recovering from Erasures

- **Erasure vs. Pauli Error:** The qubit is completely lost, but we know *where* it happened.
- **Blinded Stabilizers:** Adjacent X and Z stabilizers are randomized.
- **Super-Stabilizers:** Merge the blinded stabilizers to "cancel out" the erased edge.

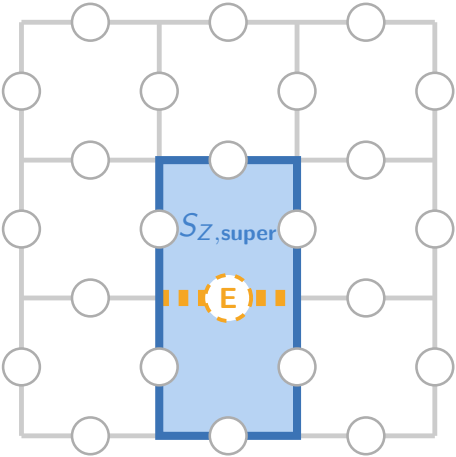
Weight-6 Super-Stabilizer



Recovering from Erasures

- **Erasure vs. Pauli Error:** The qubit is completely lost, but we know *where* it happened.
- **Blinded Stabilizers:** Adjacent X and Z stabilizers are randomized.
- **Super-Stabilizers:** Merge the blinded stabilizers to "cancel out" the erased edge.
- **Massive Threshold:** The decoder doesn't have to guess locations, it can correct erasures up to **50%** loss rate.

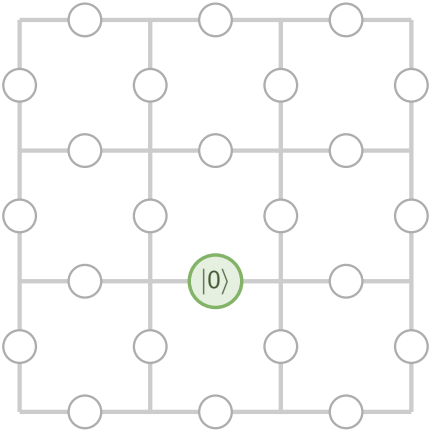
Weight-6 Super-Stabilizer



Hardware Recovery of an Erasure

1. **Injection:** A fresh $|0\rangle$ physical qubit is inserted into the erased location.

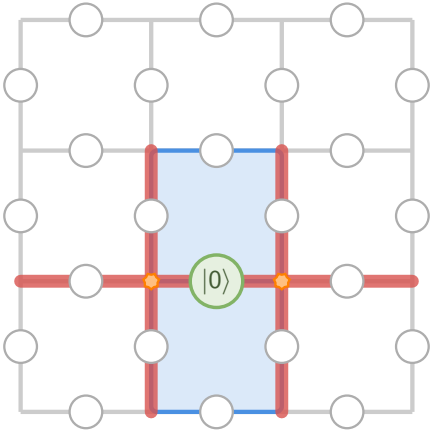
Step 1: Inject Fresh $|0\rangle$



Hardware Recovery of an Erasure

1. **Injection:** A fresh $|0\rangle$ physical qubit is inserted into the erased location.
2. **Syndrome Extraction:** We measure the individual, local stabilizers (S_1, S_2) touching this new qubit.

Step 2-3: Measure Local Stabilizers



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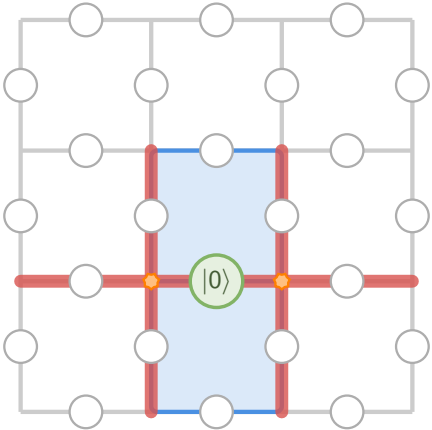

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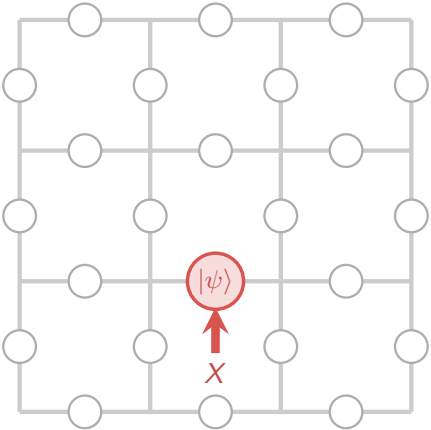
4. **State restoration:** We physically apply a Pauli X, Z, or Y gate to the new qubit to flip it into the correct logical parity.

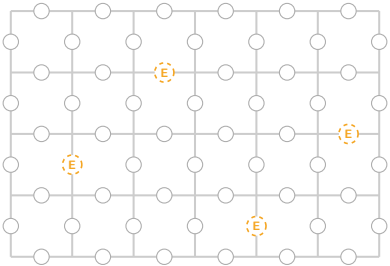
Step 4: Apply Pauli Correction



When Erasures Fail: Spanning Paths

- **The Percolation Limit:** Erasures can be fixed *unless* they span across the entire lattice.
- **The Fatal Ambiguity:** If a chain of erased qubits connects opposite boundaries, it covers a valid topological path for a Logical Operator.
- **Loss of Information:** Since all qubits on this path are missing, we cannot know if they represented the Identity (I) or a full Logical Gate (e.g., Z_L). The encoded information is permanently destroyed.

Recoverable: Isolated Erasures (Small Holes)



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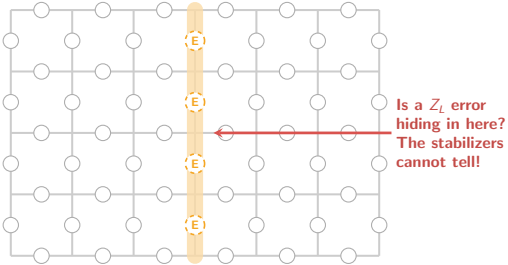
Unrecoverable: Spanning Erasure Path (Lattice is Cut)



Interactive Demo

Figure: Link to our demo